assess_cross_validation_feasibility.R

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`assess_cross_validation_feasibility`
Assess Cross-Validation Feasibility

Description

Evaluates full-model and candidate training-partition diagnostics and selects the first viable cross-validation strategy.

Usage

```
assess_cross_validation_feasibility(
  data_partition_diagnostics = NULL,
  min_train_locations = NULL,
  min_train_samples = NULL,
  min_train_taxa = NULL,
  min_mem_locations = NULL
)
```

Arguments

`data_partition_diagnostics`
: Data frame with one row for the full model and one row per candidate training partition. Required columns are ‘cv_strategy‘, ‘repeat_id‘, ‘effective_folds‘, ‘fold_id‘, ‘n_train_locations‘, ‘n_train_samples‘, ‘n_train_taxa‘, and ‘n_train_mem_locations‘. Supported strategies are ‘"full_model"‘, ‘"spatially_stratified_group_kfold"‘, and ‘"leave_one_location_out"‘.

`min_train_locations`, `min_train_samples`, `min_train_taxa`
: Positive integer thresholds required in the full model and every selected training partition.

`min_mem_locations`
: Positive integer minimum number of locations required for MEM construction.

Details

The full model is checked first. Viable grouped candidates are preferred in ascending fold-count order, followed by leave-one-location-out. A viable full model without any viable holdout requires tier-pooled regularization.

Value

One-row tibble containing full-data counts, feasibility flags, selected ‘cv_strategy‘, ‘effective_folds‘, and ‘cv_feasibility_status‘.

Examples

```
data_diagnostics <-
  tibble::tibble(
    cv_strategy = "full_model",
    repeat_id = 0L,
    effective_folds = NA_integer_,
    fold_id = 0L,
    n_train_locations = 7L,
    n_train_samples = 70L,
    n_train_taxa = 10L,
    n_train_mem_locations = 7L
  )
assess_cross_validation_feasibility(
  data_partition_diagnostics = data_diagnostics,
  min_train_locations = 5L,
  min_train_samples = 5L,
  min_train_taxa = 3L,
  min_mem_locations = 4L
)
```

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